

TB-11 EYE TRAUMA

Trauma & Burns | EMT - AEMT - Paramedic

Core principle: Treat life threats first. Irrigate chemical exposure immediately, but never apply pressure, irrigate, manipulate, instill drops, or remove an object when a penetrating or open-globe injury is suspected.

QUICK FLOW

1. SCENE + LIFE THREATS PPE, ABCs, associated head/facial trauma; protect spine only when indicated.	2. CHEMICAL / THERMAL? Immediate copious irrigation; remove contact lens only if easy; no neutralization.	3. OPEN GLOBE? No pressure, manipulation, irrigation, drops, or object removal; rigid shield.
4. BLUNT / SUPERFICIAL? Check vision and red flags; shield suspected globe injury; irrigate loose debris only.	5. VISION THREAT? Vision loss, irregular pupil, hyphema, proptosis, extrusion, severe pain.	6. PROTECT + TRANSPORT Pain/nausea control, minimize Valsalva, serial vision, notify early.

PROVIDER-LEVEL ACTIONS

EMT	Treat ABCs and associated trauma. Begin chemical irrigation immediately. For suspected open globe, do not press, manipulate, irrigate, instill medication, or remove an object; stabilize without pressure and apply a rigid shield. Obtain visual acuity in each eye when feasible without delaying care.
AEMT	All EMT actions plus clinically indicated vascular access and authorized analgesia/antiemetic care. Monitor associated head/facial trauma and support time-sensitive transport without delaying irrigation or eye protection.
PARAMEDIC	All prior actions; lead advanced airway, analgesia, antiemetic, monitoring, Medical Control, and destination decisions. Current Tennessee reference excludes ketamine analgesia in penetrating eye trauma unless specifically authorized.

FOCUSED ASSESSMENT

- Mechanism, time, laterality, product or chemical, contact-lens use, protective eyewear, high-velocity metal/glass, and associated head or facial trauma.
- Visual acuity in each eye separately before and after treatment when feasible. If no chart: document reading, finger count, hand motion, or light perception.
- Inspect without pressure: pupil shape/reactivity, redness, bleeding, hyphema, corneal clouding, foreign body, lid injury, proptosis, peri-orbital swelling, and visible tissue or fluid.
- Assess extraocular movement or diplopia only gently and without manipulating a possible open globe. Never force the eyelids open.

VISION-THREATENING FINDINGS

Urgent transport / early notification: decreased or lost vision; irregular or teardrop pupil; visible penetration or extrusion; hyphema; impaled object; marked proptosis or rapidly increasing swelling; severe pain; significant chemical burn; high-velocity mechanism; or inability to examine the eye.

CHEMICAL / THERMAL EYE INJURY

- Begin copious water or normal-saline irrigation immediately and continue for at least 15 minutes and during transport as needed. Water is preferable to waiting for saline or specialty fluid.
- Remove contact lenses only when easily possible without delaying irrigation. Irrigate from the inner corner outward and prevent runoff into the unaffected eye or onto responders.
- Brush away dry chemical before irrigation when safe and appropriate. Do not attempt acid-base neutralization. Use TE-09 Chemical / HazMat Exposure and Poison Help for unusual or water-reactive agents.
- For thermal injury, gently cool or irrigate with room-temperature water or saline. Avoid ice, ointment, and pressure.

PENETRATING / OPEN-GLOBE INJURY

- Do not remove an impaled object, manipulate the eye, apply pressure, irrigate, or instill medication.
- Stabilize a protruding object with bulky material around it without pressure. Place a rigid shield over the injured eye when available.
- Provide authorized analgesia and antiemetic care; minimize coughing, vomiting, agitation, and Valsalva. Elevate the head when safe for the overall trauma condition.
- Do not routinely patch both eyes. Covering the unaffected eye may be considered only when tolerated and when fall, anxiety, airway, neurologic, and monitoring risks are controlled.

BLUNT TRAUMA / SUPERFICIAL FOREIGN BODY

- Use a rigid shield when globe injury is possible; never use a pressure dressing. Evaluate for hyphema, irregular pupil, vision change, diplopia, restricted movement, proptosis, and associated facial/head injury.
- Loose superficial debris may be irrigated only when penetrating injury or globe rupture is not suspected.
- Do not remove embedded material or use cotton swabs, magnets, or instruments on the eye.
- Marked proptosis, rapidly increasing swelling, severe pain, decreased vision, or an afferent pupillary abnormality may indicate orbital compartment syndrome: transport emergently and notify early.

MEDICATIONS

Analgesic	Use the current pain-management pathway and provider-level standing orders. Tennessee's ketamine reference excludes penetrating eye trauma.
Antiemetic	Use the current age-appropriate standing order when nausea or vomiting threatens the injured eye.
Topical anesthetic	Only if specifically adopted or ordered by Medical Control; never use with suspected penetration/open globe and never dispense for home use.

DO NOT / DOCUMENT

- Never apply pressure to a traumatically injured eye or possible open globe. Sudden painless non-traumatic vision loss belongs in a medical/Medical Control pathway, not this trauma pathway.
- Do not delay chemical irrigation to identify the product, measure visual acuity, find saline, or make a consultation call.
- Document baseline and repeat vision, pupils, laterality, mechanism, product/exposure time, irrigation start/duration, protection, medications, response, consultation, and destination rationale.

DRAFT FOR REVIEW. Confirm stocked rigid shields and irrigation equipment, medication formulary, topical-anesthetic policy, specialty destination plan, and ophthalmology/trauma-system review before clinical release.

Reference: Tennessee EMS ALS/BLS Blended Protocol Guidelines 2024-2025, SOP #405 Eye Trauma and Ketamine Reference. Claiborne County approved standing orders and Medical Control govern final care.